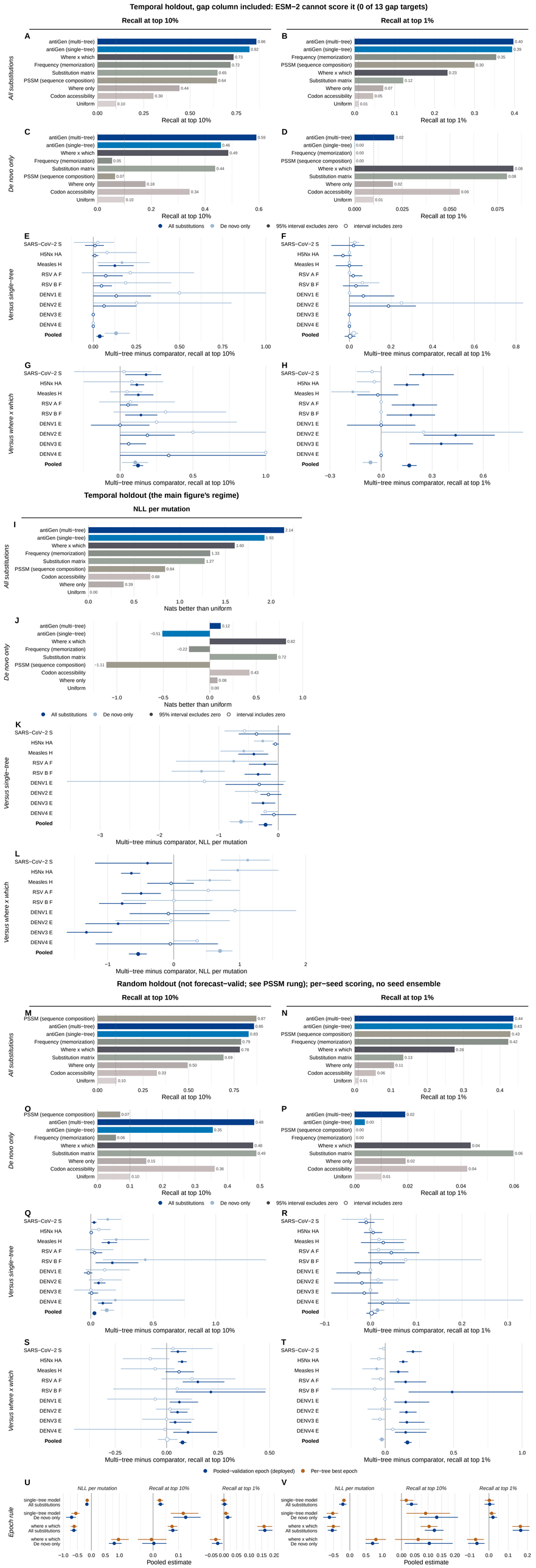
Temporal holdout, gap column included: ESM−2 cannot score it (0 of 13 gap targets)
Recall at top 10%
Recall at top 1%
All substitutions
De novo only
Versus single−tree
Versus where x which
antiGen (multi−tree)
antiGen (single−tree)
Where x which
Frequency (memorization)
Substitution matrix
PSSM (sequence composition)
Where only
Codon accessibility
Uniform
Recall at top 10%
Recall at top 1%
All substitutions
De novo only
95% interval excludes zero
interval includes zero
SARS−CoV−2 S
H5Nx HA
Measles H
RSV A F
RSV B F
DENV1 E
DENV2 E
DENV3 E
DENV4 E
Pooled
Multi−tree minus comparator, recall at top 10%
Multi−tree minus comparator, recall at top 1%
Temporal holdout (the main figure's regime)
NLL per mutation
Nats better than uniform
Multi−tree minus comparator, NLL per mutation
Random holdout (not forecast−valid; see PSSM rung); per−seed scoring, no seed ensemble
Pooled−validation epoch (deployed)
Per−tree best epoch
Epoch rule
NLL per mutation
Recall at top 10%
Recall at top 1%
single−tree model All substitutions
single−tree model De novo only
where x which All substitutions
where x which De novo only
Pooled estimate